

5.2.19

EE25BTECH11016 - Taraka Abhinav

Question: Solve the system of equations

$$x + y = 14 \quad (0.1)$$

$$x - y = 4 \quad (0.2)$$

Solution:

The equation of line:

$$\mathbf{n}^T \mathbf{x} = c \quad (0.3)$$

Line L:

$$\begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 14 \quad (0.4)$$

Line K:

$$\begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 4 \quad (0.5)$$

These can be combined and written in matrix form:

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 14 \\ 4 \end{pmatrix} \quad (0.6)$$

The following augmented matrix can be solved by gaussian elimination

$$\left(\begin{array}{cc|c} 1 & 1 & 14 \\ 1 & -1 & 4 \end{array} \right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cc|c} 1 & 1 & 14 \\ 0 & -2 & -10 \end{array} \right) \quad (0.7)$$

Next, divide the second row by -2 :

$$R_2 \leftarrow \frac{R_2}{-2} : \left(\begin{array}{cc|c} 1 & 1 & 14 \\ 0 & 1 & 5 \end{array} \right) \quad (0.8)$$

Subtract R_2 from R_1 to eliminate y :

$$R_1 \leftarrow R_1 - R_2 : \left(\begin{array}{cc|c} 1 & 0 & 9 \\ 0 & 1 & 5 \end{array} \right) \quad (0.9)$$

So, the solution is

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 9 \\ 5 \end{pmatrix} \quad (0.10)$$

